

Team D – Frontend Pages

Smart Campus Lab & Resource Optimizer

Detailed Frontend Page Specification

This document defines the pages Team D should design and develop in Phase 1. It covers only the user-facing frontend. Backend, BFF implementation, database implementation, scheduling algorithms and routing algorithms are outside this document.

Recommended main pages: 10

1. Login · 2. Dashboard · 3. Lab Search & Availability · 4. Lab Details · 5. Create Booking · 6. My Bookings · 7. Booking Details & Status · 8. Live Monitoring · 9. Campus Map / Route · 10. Reports & Admin

Page 1 — Login

Purpose: Provide the entry point to the application and establish the user's frontend experience.

Main UI elements

- Application logo/name
- Email / username input
- Password input
- Show/hide password control
- Login button
- Forgot password link, if included in project scope
- Loading indicator during login
- Validation messages
- Authentication error message
- Session-expired message

User information displayed/used

- User name
- User role: Student / Faculty / Lab Admin
- Optional department/profile information

Frontend states

- Empty form
- Invalid input
- Loading
- Login success
- Login failure
- Session expired

Navigation

Data dependency: Team C provides user/role information. In Phase 1, use mock user accounts.

Page 2 — Dashboard

Purpose: Give the user an immediate overview of campus labs, resources and their current activity.

Recommended dashboard cards

Card	Example information
Active / Available Labs	12 available · 4 occupied
My Active Bookings	2 active bookings
Pending / Queued	1 booking in queue
Available Resources	48 available resources
Alerts	2 important alerts

Dashboard sections

- Quick action: Search Lab
- Quick action: New Booking
- Upcoming bookings
- Current booking status
- Lab/resource availability summary
- Queue summary
- Recent alerts/notifications
- Last updated timestamp

Role-aware content

- Student: personal bookings, search and status
- Faculty: bookings plus relevant monitoring/report information
- Lab Admin: resource status, maintenance and operational information

Data dependency: Mainly Team C, with Team A for scheduling/queue information and Team B for route-related quick information where shown.

Page 3 — Lab Search & Availability

Purpose: Allow users to find a suitable lab based on their requirements.

Search/filter controls

- Date
- Start time
- End time
- Required capacity
- Required equipment/capability
- Location/building filter, if required
- Availability filter
- Search / Apply Filters button

Search result card

- Lab name and ID
- Location
- Capacity
- Availability status
- Available resources/equipment
- Selected date/time
- View Details button
- Book button

Important UI states

- No matching labs
- Loading results
- Search error
- Lab unavailable
- Lab available

Data dependency: Team C provides lab, resource and availability data.

Page 4 — Lab Details

Purpose: Show complete information about one selected lab before booking.

Lab information

- Lab name and ID
- Building/floor/location
- Capacity
- Current status
- Availability for selected date/time

Resources / capabilities

- Resource/equipment name
- Quantity
- Availability
- Operational / maintenance status

Actions

- Select date/time
- Select resources
- Book this lab
- View route to lab

Visual design

- Lab summary card
- Resource list/table
- Availability indicator
- Location/map preview
- Clear primary Book button

Data dependency: Team C provides lab/resource information; Team B provides route information if a route preview is shown.

Page 5 — Create Booking

Purpose: Provide a clear, validated booking workflow.

Booking form fields

- Selected lab
- Date
- Start time
- End time
- Required resources/equipment
- Purpose of booking

Frontend validation

- Required fields cannot be empty
- End time must be after start time
- Invalid date/time should be rejected
- Requested resource selections should be clearly shown
- Prevent accidental repeated submissions

Confirmation step

- Booking summary
- Lab/date/time
- Selected resources
- Purpose
- Confirm Booking / Go Back

Result states

- Submitting
- Booking accepted
- Booking queued
- Booking rejected
- Temporary error with retry

Data dependency: Team C supplies booking/lab/resource information; Team A supplies scheduling outcome/status after submission.

Page 6 — My Bookings

Purpose: Let users view and manage their bookings from one place.

Tabs/filters

- Upcoming
- Active
- Queued/Pending
- Completed
- Cancelled/Rejected
- Date filter

Booking card/table

- Booking ID
- Lab
- Date
- Start/end time
- Current status
- Purpose
- View Details button

Possible actions

- View details
- Cancel booking, if allowed
- Open route
- Refresh status

Important states

- No bookings
- Loading
- Booking list error
- Status update
- Cancelled/expired booking

Data dependency: Team C is the primary source for booking records and status.

Page 7 — Booking Details & Status

Purpose: This is the most important integration screen. It combines booking, scheduling, resource and routing information into one user-facing view.

Section 1 — Booking summary

- Booking ID
- Lab
- Date/time
- Purpose
- Current status

Section 2 — Status tracker

REQUESTED → VALIDATED → QUEUED → LEASED → CONFIRMED → ACTIVE → COMPLETED

Alternative statuses: Rejected · Waitlisted · Cancelled · Expired

Section 3 — Scheduling / queue

- Queue position
- Requests ahead
- Estimated start time
- Scheduling state

Section 4 — Allocated resources

- Resource names/IDs
- Quantity
- Current status

Section 5 — Route

- Source
- Destination
- Distance
- Estimated travel time
- Route status
- View Route button

Data dependency: Team C + Team A + Team B.

Page 8 — Live Monitoring

Purpose: Present the current operational state of labs, resources and queues.

Monitoring widgets

- Lab status cards: Available / Occupied / Maintenance
- Resource status list
- Queue information
- Current active bookings
- Alerts
- Last updated timestamp
- Refresh/reconnect indicator

Example status display

Lab	Status	Optional details
Lab 1	AVAILABLE	42/60 capacity
Lab 2	FULL	60/60 capacity
Lab 3	MAINTENANCE	Unavailable
Lab 4	AVAILABLE	18/40 capacity

Frontend states

- Live/connected
- Updating
- Stale data warning
- Connection/retrieval error
- No monitoring data

Data dependency: Team C for lab/resource state; Team A for queue/scheduling state where displayed.

Page 9 — Campus Map / Route

Purpose: Visually show how the user reaches a lab or campus resource.

Map/topology elements

- Current location/source
- Buildings/nodes
- Lab destination
- Connected paths
- Highlighted route

Route result

- Destination
- Step-by-step route
- Distance
- Estimated travel time
- Route status

Example

Current Location → Main Gate → CSE Block → First Floor → Computer Lab 3

Route failure/degraded UI

- Route available
- Route temporarily unavailable
- Route calculation failed
- Retry/recalculation message

Important: Team D presents the route visually. Team B is responsible for route calculation/topology.

Page 10 — Reports & Admin

Purpose: Provide role-appropriate operational and historical views, especially for Lab Admin.

Reports section

- Lab utilization
- Resource utilization
- Booking counts
- Usage by date
- Usage by lab
- Trend charts
- Summary cards

Admin/resource section

- Resource list
- Resource ID/type
- Lab/location
- Operational status
- Maintenance status
- Availability
- Resource details
- Status/change action, if permitted

UI components

- Tables
- Cards
- Charts
- Filters
- Date range selector
- Status badges
- Confirmation dialogs
- Success/error notifications

Data dependency: Team C supplies resource/history/report data; Team A supplies scheduling metrics if those metrics are included.

Frontend Navigation Structure

Recommended navigation

LOGIN



DASHBOARD

■■■■ LAB SEARCH & AVAILABILITY

■ ■■■■ LAB DETAILS → CREATE BOOKING

■■■■ MY BOOKINGS → BOOKING DETAILS & STATUS

■ ■■■■ VIEW ROUTE

■■■■ LIVE MONITORING

■■■■ CAMPUS MAP / ROUTE

■■■■ REPORTS & ADMIN

Do not create unnecessary separate pages for:

- Queue position — keep it inside Booking Details
- Booking status — keep it inside Booking Details
- Allocated resources — keep it inside Booking Details
- Lab availability — keep it inside Lab Search/Lab Details
- Utilization cards — keep them inside Dashboard/Reports

These are functions/components, not necessarily separate pages.

Phase 1 Development Checklist

- Design all 10 main frontend pages.
- Define common layout, navigation and role-aware menus.
- Create realistic mock data for users, labs, resources and bookings.
- Create mock scheduling data for queue/status/estimated start.
- Create mock routing data for map/route screens.
- Build complete user flow from Login to Booking Details.
- Build loading, empty, success, error and retry states.
- Make the interface responsive for desktop/tablet/mobile.
- Use accessible forms, labels, contrast and keyboard navigation.
- Prepare the frontend so mock data can later be replaced by real integrated data.

Final Phase 1 outcome: Team D should be able to demonstrate the complete frontend user journey even before Teams A, B and C are integrated.

Final Phase 2 outcome: Replace mock data with the real outputs of Teams A, B and C and perform end-to-end integration testing.